Category:

Forensic

Name:

Simple Forensic

Message:

You are provided with following evidence.

1. Full system memory dump
2. USN Journal
3. MFT file
4. Prefetch files

One of programs (processes) captured in the memory dump file contains the flag.

https://cyberseagame2025.s3.us-east-2.amazonaws.com/Resources/memdump.zip

https://cyberseagame2025.s3.us-east-2.amazonaws.com/Resources/evidence.zip

Objective:

Identify the correct process and extract the flag.

Instructions:

The memdump.zip contains an entire system memory dump with 10 remote shell programs all named “CSG2025_Forensic_Challenge.exe”.

The target process is the file created on “2025/09/10 08:18:20” running from .¥Users¥tsshimizu¥Desktop¥Tor Browser¥0010”.

(Easiest way to identify this is by looking into MFT file probably)

Using Volatility3, the target process memory dump can be extracted.

By analyzing the binary, you will find followings

- flag_string: 8I8UA7wRkZlDgKEN0EAtic6wcVhGpuW7avv3xEemfliBAYXtpgJhNukG9tm7wRod
- Encrypted AES Key: 12021e082708132b341a01182416341b063f365b21282631242340433f39311e
- Encrypted IV: 2a3c240a2532392337202700221601 0e043c185c08175158
- xorKey: simplekey

Once the encrypted flag is retrieved, all you need to do is decode the AES key and IV with xorKey and decrypt the flag.

- Decrypted Flag: aB!3dE#5fG@7hI^9jK&1lM*2nO(4pQ)6rS-8tU_0vW+z